



AGH UNIVERSITY OF SCIENCE
AND TECHNOLOGY

Function Sequence and Series for Electronics and Telecommunications

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- What is functional analysis about?
- Function sequence
- Uniform and pointwise convergence
- Function series
- Differentiation and integration of the series
- Examples

What is functional analysis?



This is area of mathematics which investigate spaces of functions. It all started with the outstanding German mathematician David Hilbert around 1904 and his collaboration with Erhard Schmidt on problems concerning integral equations. They had a breakthrough idea, as many of Hilbert's foresights, to look at the spaces in which the problem is set up, as at the vector space, so that each function would have a length assigned to it, and we could talk about the angel between two functions. In a similar way to the way we talk about length of a vector in $\mathbb{R}^3$ and the angel between two vectors in $\mathbb{R}^3$. Both notions require the scalar product.

- 1-dim space $\mathbb{R}$, elements $x \in \mathbb{R}$ (points)
- 3-dim space $\mathbb{R}^3$, elements $(x, y, z) \in \mathbb{R}^3$ (triplets)
- 4-dim space $\mathbb{R}^4$, elements $(x, y, z, t) \in \mathbb{R}^4$ (quadruples)
- n -dim space $\mathbb{R}^n$ ($n \in \mathbb{N}$), elements $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ (n -tuples)
- countably dimensional space l^2 , elements are sequences of numbers $(x_1, x_2, x_3, \dots) \in l^2$, st. $\sum_{n=1}^{\infty} x_n^2$ is finite
- uncountably dimensional space of functions $L^2(X, Y)$, elements are functions $f : X \rightarrow Y$, where $X, Y \subset \mathbb{R}$, st. $\int_X f^2(x)dx$ is finite
- many more function spaces....

In a space of numbers (eg. $\mathbb{R}$ or $\mathbb{R}^n$) we considered sequences of numbers (a_n) defined as

$$\mathbb{N} \ni n \rightarrow a_n \in \mathbb{R}, n \in \mathbb{N}.$$

In the space of functions we define a function sequence as

$$\mathbb{N} \ni n \rightarrow f_n(x), x \in [a, b].$$

Examples of function sequences $f_n(x) = x^n, g_n(x) = \sin(nx) \quad x \in \mathbb{R}$.

If we fix a point in a domain, eg. $x = \pi$ we obtain a sequence of numbers

$$g_n(\pi) = \sin(n\pi) = 0.$$

Consider a sequence of functions $f_n : [a, b] \rightarrow \mathbb{R}$.

The sequence (f_n) is **pointwise convergent** $f_n \xrightarrow{n \rightarrow \infty} f$ iff

$$\forall \varepsilon > 0 \forall x \in [a, b] \exists N = N(x, \varepsilon) \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$$

The sequence (f_n) is **uniformly convergent** $f_n \xrightarrow{n \rightarrow \infty} f$ iff

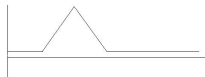
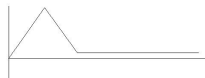
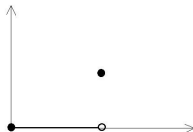
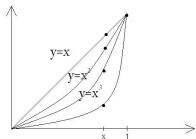
$$\forall \varepsilon > 0 \exists N = N(\varepsilon) \forall x \in [a, b] \forall n \geq N : |f_n(x) - f(x)| < \varepsilon$$

But there are many other ways to define the limit in a space of functions. In some sense those are the two extremal ways: very weak (pointwise), and very strong (uniform). Weak requirements would produce weak consequences, but strong requirements would be very hard to satisfy.



Sequences of functions pointwise convergent, **AGH** but not uniformly convergent

$$f_n(x) = x^n, f_n \in C^\infty \text{ but } f \notin C^0$$



Function series $\sum_{n=1}^{\infty} f_n(x)$

Let (f_n) , $n \in \mathbb{N}$ be a sequence of functions, which are defined at the same domain $x \in [a, b]$. Let's define partial sums of the function sequence

$$s_n = f_1 + f_2 + \dots + f_n, \text{ for any } n \in \mathbb{N}.$$

The limit of the sequence of partial sums (if it actually exists) is called the **sum of the series** and is denoted as follows

$$\lim_{n \rightarrow \infty} s_n(x) = \sum_{n=1}^{\infty} f_n(x), \quad x \in [a, b].$$

If the sequence of the partial sums is convergent/divergent, then we say that series is convergent/divergent, respectively.

Definition of Cauchy's condition

Series of functions (f_n) defined on $[a; b]$ satisfies Cauchy's condition if

$$(*) \forall \varepsilon > 0 \quad \exists T \in \mathbb{N} \quad \forall r, t \geq T(\varepsilon) \quad \forall x \in [a, b] : |f_r(x) - f_t(x)| < \varepsilon.$$

Necessary and sufficient condition of uniform convergence

Sequence (f_n) is uniformly convergent in $[a; b]$ iff it satisfies $(*)$.

For a fixed $x_0 \in [a; b]$, $(*)$ is sufficient condition of convergence $\sum f_n(x_0)$, which is a majorant of $\sum f_n$. If opposite, we assume uniform convergence $|f_n(x) - f(x)| < \varepsilon$, then Cauchy's condition follows from

$$|f_r(x) - f_t(x)| \leq |f_r(x) - f(x)| + |f(x) - f_t(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Weierstrass test

Given a series of functions

$$\sum_{n=1}^{\infty} f_n(x), \quad x \in [a, b]$$

and compounds of this series of functions are bounded by

$$|f_n(x)| \leq a_n, \text{ for all } x \in [a, b], \text{ and for almost all } n \in \mathbb{N}$$

and the number series $\sum_{n=1}^{\infty} a_n$ is convergent, then function series $\sum_{n=1}^{\infty} f_n(x)$ is uniformly convergent.

Proof of Weierstrass test of uniform convergence

Let's denote $S_n = \sum_{k=1}^n f_k(x)$ (sum of function series)

and $s_n = \sum_{k=1}^n a_k$ (sum of number series).

Then, from the assumptions $|f_n(x)| \leq a_n$ for all n and $x \in [a, b]$.

Assuming without loss of generality that $r > t$

$$|S_r(x) - S_t(x)| = |f_{t+1}(x) + f_{t+2}(x) + \dots + f_r(x)|$$

$$\leq a_{t+1} + \dots + a_r = |s_r - s_t|.$$

Since $\sum a_k$ is convergent, it fulfills Cauchy's condition ie.

$$\exists T \forall r > t > T : |s_r - s_t| < \varepsilon \text{ and also } |S_r(x) - S_t(x)| < \varepsilon$$

fulfills the Cauchy's test, so it's uniformly convergent.

Two points of view at the same function space

- **algebraic** - we look at it as at the vector space
(for two vectors (ie.functions in a function space), we define a scalar product, by which we can define length of a vector and check orthogonality; it provides a way to define a basis.)
- **analytic** - we ask about limits of sequences of points (ie. functions)
(there are many nonequivalent notions of a limit in a function space, impossible in a finitely dimensional space, also the space may be not complete ie. the limit itself may 'fell out' from the space)

There are far more problems and complications than what we haven't experienced in $\mathbb{R}$ or even $\mathbb{R}^n$.

Th.1c

If

- ① functions $f_n : [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$ for all $n \in \mathbb{N}$
 - ② $\sum_{n=1}^{\infty} f_n(x) = f(x)$ is uniformly convergent at the interval $[a, b]$
- then f is continuous at $[a, b]$.

Th.2d

If

- ① functions $f_n : [a, b] \rightarrow \mathbb{R}$ are differentiable in $[a, b]$, for all $n \in \mathbb{N}$
- ② series $\sum_{n=1}^{\infty} f_n(x) = f(x)$ is uniformly convergent at the interval $[a, b]$,

then there exists derivative of f at each point $x \in [a, b]$

$$f'(x) = \sum_{n=1}^{\infty} f'_n(x), \quad x \in [a, b].$$

So, derivative of the series is equal to the series of derivatives

$$\left(\sum f_n(x)\right)' = \sum f'_n(x).$$

Th.3i

If

- ① functions f_n are integrable in $[a, b]$, for all $n \in \mathbb{N}$
- ② series $\sum_{n=1}^{\infty} f_n(x) = f(x)$ is uniformly convergent at the interval $[a, b]$,

then

$$\underbrace{\int_a^b \left(\sum_{n=1}^{\infty} f_n(x) \right) dx}_{\text{integral of the sum of the series}} = \underbrace{\sum_{n=1}^{\infty} \int_a^b f_n(x) dx}_{\text{the sum of the series of integrals}}$$

(function series)
(number series)



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Example 1 (lack of integrability of the series, violation of Th.3i)

Th.3i fails because of lack of uniform convergence. Let's consider a series defined by partial sums $s_n(x) = nx e^{-nx^2}$ defined at $[0, 1]$.

$(s_n(x))$ is a sequence of partial sums for the series $\sum_{n=1}^{\infty} f_n$, where $f_1 = s_1$, $f_n = s_{n+1} - s_n$.

$$\underbrace{s_1(x) + [s_2(x) - s_1(x)]}_{s_2(x)} + [s_3(x) - s_2(x)] + [s_4(x) - s_3(x)] + \dots$$
$$\underbrace{\hspace{10em}}_{s_3(x)}$$
$$\underbrace{\hspace{15em}}_{s_4(x) \dots}$$

$\lim_{n \rightarrow \infty} s_n(x) = \lim_{n \rightarrow \infty} nx e^{-nx^2} = 0$, because e^{-nx^2} tends to 0 faster, then polynomial nx tends to ∞ , but the convergence is nonuniform.

Example 1 (continued)

Let's calculate LHS and RHS of the expressions from Th.3i.

LHS

$$\int_0^1 (\sum_{n=1}^{\infty} f_n(x)) dx = \int_0^1 \lim_{n \rightarrow \infty} s_n(x) dx = \int_0^1 0 dx = 0$$

RHS

$$\sum_{n=1}^{\infty} \left(\int_0^1 f_n(x) dx \right) = \lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$$

$$= \lim_{n \rightarrow \infty} \left(-\frac{1}{2} \right) \int_0^1 (-2nx) e^{-nx^2} dx = \lim_{n \rightarrow \infty} \left(-\frac{1}{2} \right) \left[e^{-nx^2} \right]_0^1$$

$$= \lim_{n \rightarrow \infty} \left(-\frac{1}{2} \right) [e^{-n} - 1] = \frac{1}{2}.$$

LHS $\neq$ RHS, so the order of integration and summing can not be changed.

$$\int_0^1 (\sum_{n=1}^{\infty} f_n(x)) dx \neq \sum_{n=1}^{\infty} \left(\int_0^1 f_n(x) dx \right)$$